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PI Name:	Rothenberg, Ellen Ph.D.		
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Human Research Program Risks:	None		
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PI Email:	evroth@its.caltech.edu	**Fax:**	FY 626-449-0756
PI Organization Type:	UNIVERSITY	**Phone:**	626-395-4992
Organization Name:	California Institute of Technology		
PI Address 1:	Division of Biology 156-29		
PI Address 2:	1200 E. California Boulevard		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91125	**Congressional District:**	26
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Contact Monitor:		**Contact Phone:**	
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The lymphocyte-based adaptive immune system is a complex cellular structure which seems to have evolved entirely since the vertebrate line diverged from all other deuterostome animals. The long-term goal of this project is to discover the molecular basis for the many interlinked biological innovations that made this system possible. In the first period of funding of this project, we have made swift progress in identifying the lower-vertebrate and related invertebrate versions of the major regulatory genes that control lymphocyte development, the genes encoding a key set of hematopoietic transcription factors. This work has revealed shifts in the number, primary structure, and expression patterns of corresponding genes, with major changes apparently between the divergence times of jawless vertebrates and cartilaginous vertebrates. In this renewal proposal, we will focus on three areas. The first is the comparison between the skate, which we have found to use a full set of lymphoid regulatory genes in a similar way to mammals, and the

Task Description:	lamprey, in which initial studies of two genes indicate substantial differences in structure and/or gene number from all jawed vertebrates. We will characterize the lamprey homologues of these transcription factor genes to test the timing of gene duplications. We will determine whether conserved DNA binding domains may be linked with different protein interaction domains. Expression studies will be done to try to explain why lampreys have no true lymphocytes. Combinations of co-expressed factors in lamprey tissues will be sought as evidence for possible cryptic lymphocyte-like cells in these animals. Our initial evidence suggests that the Spi/PU.1 gene family, a key player in blood development, may be a vertebrate innovation. We will test this and determine the nature of molecular changes that could have produced this regulatory gene. Using technical advances developed through other projects in our lab, we have assay conditions that can allow a direct functional comparison between effects of the lower-vertebrate and mammalian forms of these transcription factors in lymphocyte development. In our system, overexpression of PU.1, GATA-3, or Runx1 each has a highly distinctive set of effects on development of T-cell precursors. We will use this system to score the abilities of skate and lamprey PU.1, GATA-3, and Runx to mimic mouse factors in these systems, and in assays of B-cell and non-immune developmental alternatives. The results will forge a linkage between the structural changes in these regulatory factors during evolution and their developmental function. This task is to reveal the evolutionary mechanism that enabled the ancestors of vertebrates to develop a unique new pair of cell types, the T and B lymphocytes of the immune system. This task is important to understand the ways that novel body parts and cell types appear in evolution.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work is to obtain a deeper understanding of molecular mechanisms driving the evolutionary process by focusing on two very well-understood cell types that are known to have evolved only in vertebrates. The processes that enabled these cell types to be generated are a model for the way organisms create new structures generally. A better understanding of mechanisms giving life forms adaptibility is a benefit.
Task Progress:	This project is the continuation of research initiated under NAG2-1370 (10/1/99-9/30/02), in which lymphocyte development was used as a model to define in precise molecular terms the regulatory innovations that occur in evolution to generate a new kind of body part (cell type). The rules that can be derived for lymphocyte evolution should be applicable to the development of other kinds of specialized structures, such as vertebrate “anti-gravity systems” (musculo-skeletal reorganization). In the past year, we completed the definition of the transcription factors that control the development of the cartilaginous fish immune system and published the results. In spite of the large evolutionary distance between these animals and mammals, they have a remarkably mammal-like use of transcriptional regulators. This shows that the regulatory factors important for mammalian lymphocyte development were used in generally homologous ways as early as the last common ancestor of all animals with B and T lymphocytes. The next step is to focus completely on the lamprey, an animal with lymphocyte-like cells which nonetheless use different receptor structures and possibly different immune response strategies for defense. This animal is far more evolutionarily distant from the jawed vertebrates than is the cartilaginous fish, and we have experienced some difficulty in repeated attempts to use standard cloning techniques to isolate lamprey equivalents of some of the genes of our interest. However, we have made progress in three areas of this work. (a) We have been able to use a new technological approach to isolate lamprey lymphocytes, in spite of their lack of known surface markers, using the forward and right-angle light scatter parameters of flow cytometry. This has enabled us to sort multiple samples of pure lamprey lymphocyte-like cells, and prepare high-quality lamprey cDNA for gene expression analysis in the most directly relevant cell types. (b) We have been able to clone the lamprey PU.1 equivalent, lamprey Spi, into a retroviral vector for functional tests of its activity in direct comparison with mammalian PU.1. We have also adapted a powerful in vitro system that allows us to observe characteristic perturbation effects of PU.1 and PU.1 domain mutants as they impact the development of T and B lymphocytes and myeloid cells from mammals. This will make possible a head-to-head test of the effects of mammalian PU.1 vs. lamprey Spi that will determine whether the evolutionary drift in some non-DNA-binding domain sequences of the lamprey Spi has affected its spectrum of biological activities. (c) In an extremely interesting development, we have begun a collaboration with a group that has just cloned a new type of immune receptor gene from the lamprey. These receptor genes should be the targets of the regulatory molecules that we are studying, if in fact the regulatory factors carry out truly analogous functions in lampreys and in jawed vertebrates. This is an especially exciting opportunity because it allows us to separate the question of regulatory factor roles in lymphocyte development from the question of regulatory factor roles in controlling a particular target gene: the lamprey receptor genes appear to have actions that are homologous to those of jawed vertebrate immunoglobulins, but the molecules themselves are completely unrelated in structure. In other words, this is the perfect test case for whether the immune-system-development role of the factors we are studying is truly older than the particular receptors used for immunity. We are preparing candidate cis-regulatory sequence constructs from this new lamprey receptor gene to test whether it binds and can be transactivated by the regulatory factors we and our collaborators have cloned from the lamprey, such as Spi, Ikaros, GATA2/3, and Pax2/5/8.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
Articles in Peer-reviewed Journals	Anderson, M.K.; Pant, R.; Miracle, A.L.; Sun, X.; Luer, C.A.; Walsh, C.J.; Telfer, J.C.; Litman, G.W.; Rothenberg, E.V. "Evolutionary origins of lymphocytes: Ensembles of T cell and B cell transcriptional regulators in a cartilaginous fish." The Journal of Immunology 172(10):5851-5860, May 2004. , May-2004
Articles in Peer-reviewed Journals	Rothenberg, E.V.; Pant, R. "Origins of lymphocyte developmental programs: Transcription factor evidence." Seminars in Immunology, in press. , Dec-2004